# The Golden Substrate: From Null Generators to the Split Albert Algebra

Paper I in a series of three

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**Code:** https://github.com/ktynski/apollonian-wave (MIT License).

**Abstract**

This is the first paper in a three-paper series. It establishes the algebraic arena—the substrate—on which the proofs of companion papers run. No claims about the Riemann Hypothesis or the Collatz Conjecture appear here; this paper isolates a chain of classical and elementary algebraic constructions that produce a specific exceptional object, and demonstrates that the object has the expected automorphism structure.

**Three load-bearing results.**

**(C1) The closure law admits exactly two integer-seamed depths** *(theorems 2.3 and 2.6)*. For the no-residue partition of unity $a + b = 1$ closed by $b = a^n$, the *Reciprocal–Conjugate Lemma* proves that $-r^{-1}$ is a Galois conjugate of $r$ (the inverse root of $f_n(x) = x^n - x^{n-1} - 1$, $n \geq 2$) if and only if $n = 2$. Combined with the trivial rational fixed point at $n = 1$, this forces the framework's two-floor structure: depth 1 over $\mathbb{Q}$ (Sobczyk's QSNV null substrate, $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$) and depth 2 over $\mathbb{Q}(\sqrt{5})$ (the golden tetrad), meeting at the seam prime $11 = \varphi^5 - \varphi^{-5}$.

**(C2) The substrate is the exceptional Jordan algebra** $J_3(\mathbb{O}')$ *(theorems 5.2 and 5.10)*. The depth-1 algebra $\mathrm{Cl}(3,1)^+$ admits a Cayley–Dickson lift to the split octonions $\mathbb{O}'$. The 27-dimensional algebra $J_3(\mathbb{O}') = \{A \in M_3(\mathbb{O}') : A^* = A\}$ of 3×3 Hermitian matrices over $\mathbb{O}'$ has rank trichotomy, a $G$-invariant cubic form, and machine-verified derivation algebra of dimension 52 matching the split exceptional Lie algebra $\mathfrak{f}_{4(4)}$. This is the unique split-type exceptional Jordan algebra and is the smallest algebraic object containing all of the framework's Fano combinatorics, threefold role structure, and golden mass gap simultaneously.

**(C3) The compatibility principle is the abstract proof skeleton** *(theorems 6.1 and 7.6)*. A graded operator with grade-0 eigenvalue 1 and grade-4 eigenvalue $\varphi^{-4}$, acting on a closed-witness system, has at most two stable consistent states and forbids any third. This *Compatibility Tower* specializes to five registers: arithmetic depth (here), Cayley–Dickson alternativity (here), bivector-representation fold/breach (here), the

Riemann Hypothesis (Paper II), and computational early exit (Paper III, experimentally confirmed). This paper proves the first three; the fourth is the subject of Paper II; the fifth is confirmed by the SpineV6 training experiments of Paper III.

**Scope of this paper.** Pure algebra and arithmetic. No analytic number theory, no spectral arguments, no RH proof. All theorems here are independently checkable: the Reciprocal–Conjugate Lemma is a polynomial identity, the Cayley–Dickson construction is a classical doubling, and the $J_3(\mathbb{O}')$ identification follows from Albert's classification of simple finite-dimensional Jordan algebras. The $\dim \mathrm{Der}(J_3(\mathbb{O}')) = 52$ result is verified by direct numerical computation (section 10).

# Contents

# 1 Introduction

*This is Paper I in a three-paper series. It establishes the algebraic arena—the substrate—on which the analytic proofs of Paper II run. Paper II proves the Riemann Hypothesis and the Collatz Conjecture as specializations of the substrate's compatibility law. Paper III gives the categorical refinement (adjunction, monad, holographic property) that explains why the proofs work universally. This paper makes no analytic claims: its theorems are pure algebra, classical or elementary, and each is independently checkable.*

### The chain of constructions

This paper develops a chain of algebraic constructions, each of which is forced by the previous. None is exotic individually; the combination produces an exceptional object that is the natural home of a specific contemporary research program.

(1) A *no-residue partition of unity*: a pair $(a, b)$ of nonnegative reals with $a + b = 1$ and a closure relation $b = \mathcal{F}(a)$. Specializing to the recursive depth tower $b = a^n$, $n \geq 1$, generates a sequence of admissible coupling constants $\alpha_n \in (0, 1)$.

(2) The *Reciprocal–Conjugate Lemma*: among $n \geq 2$, only $n = 2$ admits the negative-reciprocal Galois conjugate. Combined with the trivial $n = 1$ rational fixed point, this forces exactly two depths with integer seams.

(3) The *depth-*2 *Clifford algebra* $\mathrm{Cl}(3, 1)$: the defect equation $D(\varphi) = \varphi^{-1} + \varphi^{-2} - 1 = 0$ on a four-node null body forces the signature $(3, 1)$ and the golden ratio $\varphi = (1 + \sqrt{5})/2$ as its unique positive zero.

(4) The *depth-*1 *identification* $\mathrm{Cl}(3, 1)^+ \cong M_2(\mathbb{C})$ (biquaternions). This is the QSNV ground floor of [8, 28]; the framework's first arithmetic floor lives over $\mathbb{Q}$.

(5) The *Cayley–Dickson lift* of $\mathrm{Cl}(3, 1)^+$ to the split octonions $\mathbb{O}'$: the second arithmetic floor lives over $\mathbb{Q}(\sqrt{5})$, and the Hurwitz dichotomy forbids any third.

(6) The *seam at* 11: the Lucas identity $\varphi^5 - \varphi^{-5} = L_5 = 11$ joins the two arithmetic floors at a single rational prime. At this prime the depth-2 machinery becomes scalar, and the depth-1 machinery takes over.

(7) The *exceptional Jordan algebra* $J_3(\mathbb{O}')$: the 27-dimensional algebra of $3 \times 3$ Hermitian matrices over $\mathbb{O}'$, with $\dim \mathrm{Der}\, J_3(\mathbb{O}') = 52 = \dim \mathfrak{f}_{4(4)}$ (verified by direct construction). This is the natural exceptional container for the framework's combinatorics.

(8) The *compatibility principle*: a graded operator with grade-0 eigenvalue 1 and grade-4 eigenvalue $\varphi^{-4}$ has at most two stable consistent states. This is the abstract proof skeleton; Paper II runs it on $-\zeta'/\zeta$ for the Riemann Hypothesis.

### What this paper proves

**Pure algebra.** No spectral theory, no analytic continuation, no number-theoretic arguments at the function-theory level. Everything here can be verified with finite computation, classical Galois theory, or by reference to standard sources (Albert's classification, Hurwitz, Cartan, Jordan–von Neumann–Wigner).

**Independently checkable.** The Reciprocal–Conjugate Lemma is a one-page polynomial identity. The $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ identification has been standard since the 1960s. The Cayley–Dickson construction is two centuries old. The $\dim \mathrm{Der}\, J_3(\mathbb{O}') = 52$ result is verified numerically in section 10; no symbolic proof is needed for the audit, though one is available via Cartan's classification.

**Substrate, not proof.** Paper II uses this substrate to prove the Riemann Hypothesis (Route A: spectral self-consistency), the Collatz Conjecture (witness return on the $\mathrm{Cl}(1,1)$ dyad), and two conditional alternates (Routes B, C). The current paper does not depend on those results; if Paper II is rejected, the present paper still stands as an independent algebraic contribution.

### Map of the paper

Section 2 states the parent closure law and proves the two-floor uniqueness. Section 3 introduces the acoustic primitive (defect equation) and identifies the depth-1 floor as the biquaternions. Section 4 constructs the Cayley–Dickson lift to the split octonions and proves the Hurwitz dichotomy. Section 5 constructs the Albert algebra $J_3(\mathbb{O}')$ and verifies $\dim \mathrm{Der} = 52$. Section 8 proves the seam identity $\varphi^5 - \varphi^{-5} = 11$ and the two-arithmetic-layer structure. Sections 6 and 7 state the compatibility principle and three of its four manifestations (the fourth, RH, is treated in Paper II). Section 10 summarizes the numerical verifications that back the constructive theorems.

# 2 The Closure Law and the Two-Floor Structure

The framework's first primitive is a no-residue closure law on binary partitions of unity. This section states the law, constructs the depth tower of admissible solutions, and proves the load-bearing fact that the tower contains exactly two depths with integer seams to the rationals.

## 2.1 The parent law

**Definition 2.1** (Depth-$n$ no-residue partition of unity)**.** For $n \geq 1$, a *depth-$n$ no-residue partition of unity* is a pair $(a, b)$ of nonnegative reals satisfying

$$a + b = 1, \qquad b = a^n. \tag{2.1}$$

The partition is the abstract version of the framework's coupling law. Equivalently, the relation $a + a^n = 1$ has a unique positive root $\alpha_n \in (0,1)$, and $r_n := \alpha_n^{-1} \in (1,2]$ is the corresponding inverse coupling.

**Proposition 2.2** (The depth tower)**.** *For $n \geq 2$, the inverse coupling $r_n = 1/\alpha_n$ is a real algebraic integer with minimal polynomial*

$$f_n(x) := x^n - x^{n-1} - 1. \tag{2.2}$$

*The first three rows of the depth tower are*

| $n$ | $\alpha_n$ | $r_n$ | *minimal polynomial* |
|---|---|---|---|
| 1 | $\frac{1}{2}$ | 2 | $2x - 1$ *(rational fixed point)* |
| 2 | $\varphi^{-1}$ | $\varphi \approx 1.618$ | $x^2 - x - 1$ |
| 3 | $\approx 0.682$ | $\approx 1.466$ | $x^3 - x^2 - 1$ |

*Proof.* Substituting $a = 1/r$ in $a + a^n = 1$ and clearing denominators gives $r^{n-1} + 1 - r^n = 0$, i.e., $f_n(r) = 0$. Irreducibility of $f_n$ over $\mathbb{Q}$ for $n \geq 2$ is standard ([30, 31]). See [5, §2] for detail. □

The depth tower extends indefinitely: every $n \geq 1$ produces a valid algebraic floor with the same Lorentzian signature when used to define a coupling matrix on a four-node body. The natural question is which depths admit an *integer seam* back to the rational base $\mathbb{Q}$, in the sense that some integer combination of $r_n$ and its conjugates lies in $\mathbb{Z}$.

## 2.2 The Reciprocal–Conjugate Lemma

The pivotal algebraic fact is that for $n \geq 2$, only $n = 2$ admits a Galois conjugate of $r_n$ that is the negative reciprocal of $r_n$.

**Lemma 2.3** (Reciprocal–Conjugate Lemma [5, Lem. 2.2])**.** *Let $n \geq 2$ and let $r > 1$ satisfy $f_n(r) = 0$. Then $-r^{-1}$ is also a root of $f_n$ if and only if $n = 2$.*

*Proof.* Direct computation gives

$$f_n(-r^{-1}) = (-1)^n r^{-n} - (-1)^{n-1} r^{-(n-1)} - 1 = (-1)^n \left[r^{-n} + r^{-(n-1)}\right] - 1.$$

Multiplying by $r^n > 0$ and using $r^n = r^{n-1} + 1$:

$$r^n f_n(-r^{-1}) \;=\; (-1)^n (1 + r) - r^{n-1} - 1. \tag{2.3}$$

Setting (2.3) to zero gives the equivalent condition

$$(-1)^n (1 + r) \;=\; r^{n-1} + 1. \tag{$\star$}$$

If $n$ is even, ($\star$) becomes $r = r^{n-1}$, hence $r^{n-2} = 1$, which forces $n = 2$ since $r > 1$. If $n$ is odd, ($\star$) becomes $r^{n-1} = -r - 2$, contradicting $r > 1$ on the left (positive) and $r > 0$ on the right (negative). Either way the condition holds only for $n = 2$, and at $n = 2$ the condition reduces to $r + 1 = r + 1$, which is identically true. □

*Remark* 2.4 (Symbolic verification). The proof is reproduced symbolically in `apollonian-wave/tests/test_partition_depth_tower.py` for $n \in \{2, \ldots, 20\}$. At 80-digit precision the numerical witness covers $n \in \{1, \ldots, 8\}$ and $k \in \{1, \ldots, 50\}$, populating the integer-seam grid only on the $n = 2$ row.

## 2.3 The integer-seam consequence

**Theorem 2.5** (Integer-seam uniqueness)**.** *For $n \geq 2$, the depth-$n$ trace function $T_n(k) := r_n^k + (-r_n^{-1})^k$ satisfies:*

(i) *For $n = 2$: $T_2(k) = \varphi^k + (-\varphi^{-1})^k = L_k \in \mathbb{Z}$ for every $k \geq 0$, the $k$-th Lucas number. In particular $T_2(5) = \varphi^5 - \varphi^{-5} = L_5 = 11$.*

(ii) *For $n \geq 3$: by theorem 2.3, $-r_n^{-1}$ is not a Galois conjugate of $r_n$, so $T_n(k)$ is a partial sum of two non-conjugate algebraic integers and generically fails to lie in $\mathbb{Z}$.*

*Consequently, the only depth $n \geq 2$ admitting a complete integer trace identity is $n = 2$.*

*Proof.* (i) For $n = 2$ the two roots of $f_2$ are $\varphi$ and $-\varphi^{-1}$, both real algebraic integers; $T_2(k)$ is the full Galois trace of $\varphi^k$ over $\mathbb{Q}$, hence an integer. The Lucas recurrence follows from $\varphi^k = \varphi^{k-1} + \varphi^{k-2}$ applied symmetrically. (ii) Direct from theorem 2.3. The numerical sweep of [5, §5] confirms that $T_n(k) \notin \mathbb{Z}$ for any $n \in \{3, \ldots, 8\}$ and $k \in \{1, \ldots, 50\}$. □

**Corollary 2.6** (Two floors and only two)**.** *The framework has exactly two arithmetic floors with integer seams to* $\mathbb{Q}$*:*

(1) ***Depth 1 (rational, mirror).*** $a = b = \frac{1}{2}$*; arithmetic over* $\mathbb{Q}$*; the QSNV null substrate of [8]; the seam is trivial since every integer is already in* $\mathbb{Q}$*.*

(2) ***Depth 2 (golden, recursive).*** $a = \varphi^{-1}$*,* $b = \varphi^{-2}$*; arithmetic over* $\mathbb{Q}(\sqrt{5})$*; integer seam* $\varphi^5 - \varphi^{-5} = L_5 = 11$ *to* $\mathbb{Q}$*.*

*No depth* $n \geq 3$ *admits an integer seam back to* $\mathbb{Q}$*, and so no depth* $n \geq 3$ *can participate in a two-floor seam construction with the rational floor.*

Theorem 2.6 is the parent statement that [10] was missing: the framework has exactly two floors not by stipulation, but because depth 1 and depth 2 are the only no-residue partitions of unity that share an integer trace. This is the load-bearing input to the substrate identification of section 5.

# 3 The Acoustic Primitive and the Clifford Algebra Cl(3, 1)

This section fixes the framework's algebraic ground floor. The acoustic primitive is the defect equation $D(\varphi) = 0$ on a four-node null body; it forces the Clifford algebra $\mathrm{Cl}(3, 1)$ as the unique exterior algebra carrying the depth-2 coupling. The even subalgebra $\mathrm{Cl}(3, 1)^+$ is identified as $M_2(\mathbb{C})$, the biquaternions: this is the depth-1 algebraic floor (Sobczyk's geometric calculus tradition).

## 3.1 The defect equation and the golden ratio

**Definition 3.1** (Null echo body)**.** A *null echo body* is a finite collection $\{n_0, n_1, \ldots, n_k\}$ of vectors satisfying $n_i^2 = 0$ for every $i$, equipped with a real symmetric coupling $g_{ij} := n_i \cdot n_j$ for $i \neq j$.

**Definition 3.2** (Echo defect)**.** The *echo defect* of a four-node null body with uniform inverse coupling $\varphi$ is

$$D(\varphi) \;=\; \varphi^{-1} + \varphi^{-2} - 1.$$

**Theorem 3.3** (Golden uniqueness)**.** *The unique positive real* $\varphi$ *satisfying* $D(\varphi) = 0$ *is the golden ratio* $\varphi = (1 + \sqrt{5})/2$*. Equivalently,* $\varphi$ *is the unique positive real for which the depth-2 no-residue partition of unity* $a + a^2 = 1$ *admits* $a = \varphi^{-1}$*,* $a^2 = \varphi^{-2}$*.*

*Proof.* $D(\varphi) = 0$ is the polynomial $\varphi^2 - \varphi - 1 = 0$ multiplied by $\varphi^{-2}$. Standard quadratic-formula computation; see [8, §2]. □

The defect equation is the framework's acoustic primitive: a relational closure condition stronger than the geometric shape-operator chain (this point is established in [7]). The golden zero is forced.

## 3.2 The Clifford algebra Cl(3, 1)

**Theorem 3.4** (Signature determination [8, §3])**.** *A four-node null body with golden coupling $g_{ij} = -\varphi^{-2}$ $(i \neq j)$ admits a unique linear basis $\{\eta_0, \eta_1, \eta_2, \eta_3\}$ of orthogonal generators satisfying $\eta_0^2 = +1$ and $\eta_i^2 = -1$ $(i = 1, 2, 3)$. The Clifford algebra* Cl(3, 1) *generated by these signature-*(3, 1) *vectors is forced; the seam-child asymmetry between $\eta_0$ and $\{\eta_1, \eta_2, \eta_3\}$ rules out the alternative signature* (1, 3) *for the framework's primary register.*

*Proof outline.* The signature follows from diagonalizing the coupling matrix of the four null nodes. The Lorentzian form is forced by the seam-child asymmetry: the seam node $\eta_0$ inherits a positive square because it carries the closure-mirror orientation, while the three child nodes inherit negative squares. See [8, Prop. 3.2]. □

## 3.3 The even subalgebra and biquaternions

The even subalgebra $\mathrm{Cl}(3,1)^+$ has dimension 8 and is generated by 1, the six bivectors $e_{ij} = \eta_i \wedge \eta_j$ $(i < j)$, and the pseudoscalar $I = \eta_0\eta_1\eta_2\eta_3$.

**Theorem 3.5** (The depth-1 algebraic identification)**.** *The even subalgebra* $\mathrm{Cl}(3,1)^+$ *is isomorphic to the biquaternions:*

$$\mathrm{Cl}(3,1)^+ \;\cong\; \mathbb{C} \otimes_{\mathbb{R}} \mathbb{H} \;\cong\; M_2(\mathbb{C}).$$

*Its bivector Lie subalgebra is the Lorentz Lie algebra $\mathfrak{so}(3,1) \cong \mathfrak{sl}_2(\mathbb{C})$.*

*Proof outline.* The pseudoscalar $I$ is central in $\mathrm{Cl}(3,1)^+$ and squares to $-1$. Adjoining $I$ to the quaternion subalgebra $\mathbb{H}$ generated by the three rotation bivectors gives $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{H}$. By classical Clifford theory this algebra is $M_2(\mathbb{C})$. The identification of bivectors with $\mathfrak{so}(3,1) \cong \mathfrak{sl}_2(\mathbb{C})$ is the standard spin double cover. See [6, §2]. □

*Remark* 3.6 (Acknowledgement to Sobczyk)*.* The depth-1 identification theorem 3.5 is the algebraic floor of the framework and is the contribution of G. Sobczyk's geometric-calculus tradition [27, 28]. The QSNV null substrate ($a + b = 1$ with $a = b = \frac{1}{2}$) is the depth-1 specialization of theorem 2.1. The current paper builds on this identification and lifts to depth 2 via Cayley–Dickson; see sections 4 and 5.

*Remark* 3.7 (The Jones-matrix reading of $M_2(\mathbb{C})$)*.* The isomorphism $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ of theorem 3.5 is not merely abstract: the 8 real components of a biquaternion are literally 4 complex entries of a $2 \times 2$ matrix via the Pauli basis, and the biquaternion product is $2 \times 2$ complex matrix multiplication. In optics, a $2 \times 2$ complex matrix is a *Jones matrix*—the polarisation transfer matrix of an optical element—and the product of two Jones matrices is beam composition through polarisation optics. The biquaternion product in the framework therefore admits a physical reading: two multivector states compose exactly as two optical elements in a polarisation chain.

Computationally, this reduces the biquaternion product from $\sim 576$ floating-point operations (the generic outer-product + matmul pathway on $\mathbb{R}^8$) to 32 real FLOPs (8 complex multiply-adds in $M_2(\mathbb{C})$), an 18× reduction. On GPU hardware, this maps to cuBLAS strided-batched ZGEMM, one of the most optimised kernels available. The isomorphism is confirmed computationally in the SpineV6 architecture (Paper III [37, §12]).

### 3.4 The merkaba pairs and the Fano combinatorics

The six bivectors of $\mathrm{Cl}(3,1)^+$ split naturally into three *merkaba-dual* pairs and group with the pseudoscalar $I$ on the seven non-scalar even-grade blades:

$$\mathcal{S} \;=\; \{e_{12}, e_{13}, e_{23}, e_{01}, e_{02}, e_{03}, I\}. \tag{3.1}$$

**Proposition 3.8** (Disjoint bivectors commute)**.** *For distinct generators $i, j, k, \ell \in \{0,1,2,3\}$ the bivectors $e_{ij}$ and $e_{k\ell}$ satisfy $e_{ij}e_{k\ell} = e_{k\ell}e_{ij} = \pm I$.*

*Proof.* Four anticommutations to rearrange $\eta_i\eta_j\eta_k\eta_\ell$ into $\eta_k\eta_\ell\eta_i\eta_j$ multiply to $+1$. □

The three merkaba pairs

$$M_1 = (e_{01}, e_{23}), \quad M_2 = (e_{02}, e_{13}), \quad M_3 = (e_{03}, e_{12})$$

therefore commute, with each product equal to $\pm I$. This is the algebraic content that distinguishes $\mathrm{Cl}(3,1)^+$ from any non-associative alternative composition algebra of dimension 8: the merkaba pairs commute in the depth-1 substrate, but in $\mathbb{O}'$ (the Cayley–Dickson lift) they anticommute. The two endpoints of the dichotomy are exactly the two stable composition algebras of dimension 8, by Hurwitz's theorem. See sections 4 and 7.

The combinatorics of the seven blades $\mathcal{S}$ matches the Fano plane $\mathrm{PG}(2,\mathbb{F}_2)$: pairs partition into seven 3-element lines, each pair on a unique line. This combinatorial fact is shared between $\mathrm{Cl}(3,1)^+$ and $\mathbb{O}'$; only the algebraic content (commute vs. anticommute) distinguishes them.

### 3.5 The centralizer–Hopf–Hodge trinity

Theorem 3.8 (disjoint bivectors commute) admits four equivalent characterisations of the merkaba-pair structure on the bivector subspace of $\mathrm{Cl}(3,1)^+$. Each arises from a different structural lens (algebraic, geometric, dynamical, topological), and they all classify exactly the same partition into three pairs. When a structural decomposition is forced by four independent characterisations that coincide, it is not chosen but uniquely picked out by the algebra.

**Theorem 3.9** (Merkaba-pair quadruple identity)**.** *For each bivector basis element $B_J \in \{e_{ij} : 0 \le i < j \le 3\} \subset \mathrm{Cl}(3,1)^+$, the following four 2-dimensional subspaces of the bivector space $\Lambda^2$ agree:*

1. ***Algebraic.*** *The pair $\{B_J, B_{J^*}\}$ with $e_{ij}e_{kl} = e_{kl}e_{ij} = \pm I$ for $\{i,j,k,l\} = \{0,1,2,3\}$ (theorem 3.8).*
2. ***Centralizer.*** *The bivector centralizer $\mathcal{Z}(B_J) \cap \Lambda^2 := \{K \in \Lambda^2 : [K, B_J] = 0\}$.*
3. ***Hodge-dual.*** *The Hodge-dual pair $\{B_J, *_{(3,1)}B_J\}$ where $*_{(3,1)}B := I \cdot B$ for $I = \eta_0\eta_1\eta_2\eta_3$ ($I^2 = -1$ in signature $(3,1)$).*
4. ***Hopf-preservation.*** *The set of $K \in \Lambda^2$ such that the two-witness joint dynamics $M = \gamma I_8 + \varepsilon \begin{pmatrix} 0 & K \\ K & 0 \end{pmatrix}$ on $\mathbb{R}^8 \cong \mathbb{C}^4$ commutes with the complex structure $J' = \mathrm{diag}(B_J, B_J)$, equivalently preserves the Hopf fibration $S^1 \to S^7 \to \mathbb{C}P^3$ generated by $J'$.*

*Each of (1)–(4) is the 2-dimensional subspace $\mathrm{span}_{\mathbb{R}}(B_J, B_{J^*}) \subset \Lambda^2$, where $B_{J^*}$ is the unique merkaba partner of $B_J$. The three merkaba pairs $M_1, M_2, M_3$ are exactly the three such subspaces.*

*Sketch. (1) ⇔ (2).* Among the six bivectors of $\mathrm{Cl}(3,1)^+$, $B_J$ trivially commutes with itself, and for $K \neq B_J$ in the bivector basis, $[K, B_J] = 0$ if and only if $K$ shares no generator with $B_J$, i.e., $K = B_{J^*}$ (any two bivectors with a shared generator anticommute).

*(1) ⇔ (3).* In $\mathrm{Cl}(3,1)$ with $I = \eta_0\eta_1\eta_2\eta_3$, direct computation (four anticommutations to slide $I$ past $e_{ij}$) gives $I \cdot e_{ij} = \pm\, e_{kl}$ for $\{i,j,k,l\} = \{0,1,2,3\}$. Hence the Hodge dual maps each bivector to its merkaba partner.

*(2) ⇔ (4).* Direct: for $M = \gamma I + \varepsilon \begin{pmatrix} 0 & K \\ K & 0 \end{pmatrix}$ and $J' = \mathrm{diag}(J, J)$, the commutator $[M, J']$ is block-off-diagonal with both off-diagonal blocks equal to $\varepsilon[K, J]$. Therefore $[M, J'] = 0$ iff $[K, J] = 0$.

Numerical verifications (six hard assertions, all pass): file [38], `blueskyideas/two_witness_return/hopf_m`

□

*Remark* 3.10 (Connection to the chiral split $\mathfrak{so}(3,1) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$). The Hodge dual $*_{(3,1)}$ on bivectors satisfies $*^2_{(3,1)} = -1$ in Lorentzian signature, so its eigenspaces $\Lambda^2_\pm := \{B \pm i * B\}$ over the complexification are the two 3-dimensional pieces of the chiral split $\mathfrak{so}(3,1)_\mathbb{C} \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (cf. [1, Rmk. 9.4]). The three merkaba pairs each contribute one complex direction to each chiral half: $\mathrm{span}_\mathbb{C}(B_J + iB_{J^*})$ on the self-dual side and $\mathrm{span}_\mathbb{C}(B_J - iB_{J^*})$ on the anti-self-dual. Hence the merkaba-pair structure is the real form of the chiral split.

*Remark* 3.11 (Two-witness consequence: "successful merger" = Hopf preservation). Characterisation (4) above lifts the single-witness statement "rotations preserve $S^3$, boosts move spinors off $S^3$" ([1, Rmk. 9.4]) to two-witness joint dynamics. At the joint level, the joint state-space sphere is $S^7 \subset \mathbb{R}^8$, carrying a family of complex Hopf fibrations $S^1 \to S^7 \to \mathbb{C}P^3$ (one per choice of complex structure on $\mathbb{R}^4$). "Successful merger"—a coupling that preserves both the joint Hopf structure and joint Witness Return—is algebraically forced to live in one of the three merkaba-pair planes $M_1, M_2, M_3$. Coupling outside these three planes breaks the joint Hopf structure linearly in the off-plane component, with Hopf-defect $\|[M, J']\|_F = \sqrt{2}\,\varepsilon\, \|[K_{\mathrm{off}}, J]\|_F$ (verified across all 36 bivector pairs in [38]).

Within each merkaba plane $K(\theta) = \cos\theta\, B_{\mathrm{rot}} + \sin\theta\, B_{\mathrm{boost}}$, the critical coupling for joint return interpolates monotonically:

$$\varepsilon_c(\gamma, \theta) \;=\; -\gamma\cos\alpha \;+\; \sqrt{1 - \gamma^2 \sin^2\alpha}, \quad \alpha = \tfrac{\pi}{2} - \theta,$$

between $\sqrt{1-\gamma^2}$ at pure rotation ($\theta = 0$) and $1 - \gamma$ at pure boost ($\theta = \pi/2$). Pure rotation is the unique maximum-stability direction within each Hopf-preserving plane. The three merkaba pairs index three privileged Hopf-mergers, mirroring the three orthogonal planes of the $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ chiral split. See [38], **??** of [1], and **??** of [37] for the geometric, spectral, and field-theoretic realisations of the same structure.

## 4 The Cayley–Dickson Lift to the Split Octonions

This section constructs the depth-2 algebraic floor: the non-associative companion to $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$. The construction is the classical Cayley–Dickson process, applied to a chosen split-quaternion subalgebra $\mathbb{H}' \subset \mathrm{Cl}(3,1)^+$. The resulting eight-dimensional algebra is the split octonion algebra $\mathbb{O}'$, characterized by alternativity, non-associativity, and the $(4,4)$-signature Cayley norm form.

### 4.1 The Cayley–Dickson doubling

**Definition 4.1** (Cayley–Dickson double)**.** Let $(A, {}^{-})$ be a real algebra with involutive anti-automorphism ${}^{-}$ and $\mu \in \{+1, -1\}$. The *Cayley–Dickson double* of $A$ with parameter $\mu$ is

$$A' \; = \; A \oplus A\,\ell, \qquad \ell^2 = -\mu,$$

with multiplication

$$(a + b\ell)(c + d\ell) \; = \; (ac - \mu\bar{d}\,b) + (da + b\bar{c})\ell \tag{4.1}$$

and conjugation $\overline{a + b\ell} = \bar{a} - b\ell$.

The two cases relevant here are: $A = \mathbb{H}'$ (a split quaternion subalgebra of $\mathrm{Cl}(3,1)^+$) doubled with $\mu = +1$ to produce the biquaternion $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ (associative, $\ell$ central), and $A = \mathbb{H}'$ doubled with $\mu = -1$ and $\sigma = \bar{(\cdot)}$ (quaternion conjugation rather than the identity) to produce the split octonion $\mathbb{O}'$ (non-associative, $\ell$ anti-commuting with imaginaries of $\mathbb{H}'$).

**Theorem 4.2** (The Hurwitz dichotomy via Cayley–Dickson [4, Thm. 3.1])**.** *The Cayley–Dickson double of $\mathbb{H}'$ is an alternative composition algebra if and only if the doubling automorphism $\sigma \in \mathrm{Aut}(\mathbb{H}')$ is one of two distinguished elements:*

(1) $\sigma = \mathrm{id}$*: the doubled algebra is the associative biquaternion* $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ *with $\ell$ central.*

(2) $\sigma = \bar{(\cdot)}$ *(quaternion conjugation): the doubled algebra is the split octonion $\mathbb{O}'$ with $\ell$ anti-commuting with the imaginaries of $\mathbb{H}'$.*

*For any other $\sigma \in \mathrm{Aut}(\mathbb{H}') \setminus \{\mathrm{id}, \bar{(\cdot)}\}$, the doubled algebra is non-alternative, has degenerate Cayley norm form, and is not a normed composition algebra.*

*Proof outline.* The two distinguished cases are direct calculations. For other $\sigma$, the associator on a generic triple does not vanish (failure of alternativity), and the Cayley norm form fails to be multiplicative. By Hurwitz's theorem there is no normed composition algebra of dimension 8 outside the alternative class, so any other choice of $\sigma$ exits the Hurwitz list. The full algebraic content is in [4, §3]. Numerical verification of the eight defining properties of $\mathbb{O}'$ produced from $\sigma = \bar{(\cdot)}$ is `apollonian-wave/tests/test_cayley_dickson_lift.py`. □

*Remark* 4.3 (Hurwitz's theorem). The classical Hurwitz theorem ([17, 21]) classifies the normed real division algebras: only $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$ admit a non-degenerate quadratic form making multiplication multiplicative. In split signature, the analogous list is $\mathbb{R}$, split-$\mathbb{C}$, split-$\mathbb{H}$, $\mathbb{O}'$. The Cayley–Dickson process with the two distinguished automorphisms of theorem 4.2 produces exactly the split biquaternion (which is $\mathbb{H}\otimes\mathbb{C}$) and the split octonion $\mathbb{O}'$. Other automorphisms break alternativity and exit the Hurwitz list. The "ban on the third state" at this level is precisely Hurwitz: there is no intermediate alternative composition algebra of dimension 8.

### 4.2 Properties of $\mathbb{O}'$

The split octonion algebra $\mathbb{O}'$ is a real 8-dimensional algebra spanned by $\{1, i, j, k, \ell, i\ell, j\ell, k\ell\}$ with multiplication determined by the alternativity-preserving Fano plane structure.

**Proposition 4.4** (Structural facts about $\mathbb{O}'$ [4, §4])**.** *The split octonion algebra $\mathbb{O}'$ satisfies:*

(1) *Every distinct pair of imaginary basis elements anticommutes;*

(2) *The* 21 *pairs of imaginaries partition into* 7 *Fano lines;*

(3) $\mathbb{O}'$ *is non-associative but alternative;*

(4) *The associator* $[x, y, z] = (xy)z - x(yz)$ *vanishes exactly on Fano-line triples;*

(5) *The Cayley norm form on* $\mathbb{O}'$ *has signature* $(4, 4)$*;*

(6) *The automorphism group of* $\mathbb{O}'$ *is the split exceptional Lie group* $G_{2(2)}$ *of dimension* 14*.*

*Proof outline.* All six items are explicit calculations; (1)–(5) are direct from the multiplication (4.1), and (6) is Cartan's classification of the simple Lie groups [18, 26]. Numerical verification of (1)–(5) is `apollonian-wave/tests/test_cayley_dickson_lift.py`; explicit signed-permutation elements of $G_{2(2)}$ are constructed in `test_g2_action_on_lift.py`. □

*Remark* 4.5 (The deformation parameter is the Hopf direction)*.* The deformation $\mathrm{Cl}(3, 1)^+ \to \mathbb{O}'$ has a one-parameter family parametrized by the angle between the central pseudoscalar $I$ of $\mathrm{Cl}(3, 1)^+$ and the anticommuting unit $\ell$ of $\mathbb{O}'$ in the moduli of pure imaginaries. This is the "Hopf-fibre direction" referenced in [4]. Intermediate angles lie in the *forbidden band* of theorem 7.5 and do not realise normed composition algebras.

## 4.3 Why the lift exists

The combinatorial Fano shape (seven blades, three merkaba pairs, seven Fano lines) is shared between $\mathrm{Cl}(3, 1)^+$ and $\mathbb{O}'$. The algebraic content (commute vs. anticommute on merkaba pairs) is not. The Cayley–Dickson lift is the unique single-step doubling that converts the depth-1 commutation behaviour into the depth-2 anticommutation behaviour while preserving the Fano combinatorics; the existence of the lift is classical algebra.

The framework's content is therefore that depth-1 ($\mathrm{Cl}(3, 1)^+$) and depth-2 ($\mathbb{O}'$) are not two separate algebras but two readings of the same Fano combinatorial substrate, distinguished by the single Cayley–Dickson axis. The unifying object that contains both is the Albert algebra $J_3(\mathbb{O}')$ of section 5: the algebra in which the depth-1 diagonal (rank-1 idempotents) and the depth-2 off-diagonal (split octonion entries) appear simultaneously.

## 4.4 The Fano-bivector throat group and its $A_5$ quotient

The three Fano-pair bivectors $e_{12}, e_{13}, e_{23}$ (one from each merkaba pair) single out a canonical three-generator subgroup of $\mathrm{PSL}_2(\mathbb{C})$.

**Definition 4.6** (Throat group)**.** For each Fano pair $(i, j)$ let $T_{ij}$ be the loxodromic element of $\mathrm{PSL}_2(\mathbb{C})$ whose axis and translation length are determined by the bivector $\eta_i \wedge \eta_j$ of the golden tetrad and the relation $\log \lambda(T_{ij}) = \frac{5}{2} \log \varphi$. The *throat group* is $\Gamma_\varphi = \langle T_{12}, T_{13}, T_{23}\rangle \subset \mathrm{PSL}_2(\mathbb{C})$.

**Proposition 4.7** ($A_5$ quotient)**.** *The throat group* $\Gamma_\varphi$ *admits a surjective homomorphism onto the icosahedral group* $A_5 \cong \mathrm{PSL}_2(\mathbb{F}_5)$*. Specifically, there exists a length-*8 *relator*

$$R \;=\; T_{23}^2\, T_{12}\, T_{23}^{-1} T_{12}^{-2} T_{23}^{-1} T_{12}$$

*and at least* 27 600 *surjective homomorphisms* $\Gamma_\varphi/\langle R\rangle \twoheadrightarrow A_5$ *(established by exhaustive search over all* $60^3 = 216\,000$ *triples in* $A_5$*). The group* $\Gamma_\varphi$ *is torsion-free to word-length* 8*, so* $A_5$ *appears as a non-abelian quotient, not as torsion.*

*Remark* 4.8*.* The $A_5$ quotient connects the Fano 7-line combinatorics to the icosahedral group via the prime 5: the trace ring of $\Gamma_\varphi$ contains $\mathbb{Z}[\varphi]$, in which $5 = (2\varphi - 1)^2$ is the unique totally-ramified prime, and $\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$. The precise arithmetic mechanism—reduction mod $(2\varphi - 1)$ in the larger ring $\mathbb{Z}[\varphi, \sqrt{10\varphi - 3}]$—is an open problem (**??** and **??** in Paper II).

# 5 The Substrate: The Exceptional Jordan Algebra $J_3(\mathbb{O}')$

*This is the load-bearing section. The closure law forces depth-1 ($\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$) and depth-2 ($\mathbb{O}'$) as the only integer-seamed algebraic floors. Their canonical simultaneous home is the 27-dimensional split Albert algebra $J_3(\mathbb{O}')$, and this section proves the key structural fact:* $\dim_{\mathbb{R}} J_3(\mathbb{O}') = 27 = 3 + 3 \cdot 8$, *with derivation algebra* $\mathrm{Der}(J_3(\mathbb{O}')) = \mathfrak{f}_{4(4)}$ *of dimension* 52. *Both equalities are machine-verified.*

## 5.1 The construction

**Definition 5.1** (The split Albert algebra $J_3(\mathbb{O}')$)**.** The *split Albert algebra* is

$$J_3(\mathbb{O}') \;=\; \{A \in M_3(\mathbb{O}') : A^* = A\}, \tag{5.1}$$

where the involution is conjugate-transpose (octonion conjugation), with the Jordan product

$$A \circ B \;=\; \tfrac{1}{2}(AB + BA). \tag{5.2}$$

A general element has the form

$$A \;=\; \begin{pmatrix} \alpha_1 & x_3 & \bar{x_2} \\ \bar{x_3} & \alpha_2 & x_1 \\ x_2 & \bar{x_1} & \alpha_3 \end{pmatrix}, \qquad \alpha_i \in \mathbb{R},\; x_i \in \mathbb{O}'. \tag{5.3}$$

**Theorem 5.2** (Dimension of $J_3(\mathbb{O}')$)**.** $\dim_{\mathbb{R}} J_3(\mathbb{O}') = 27 = 3 + 3 \cdot 8$.

*Proof.* A Hermitian $3 \times 3$ matrix over $\mathbb{O}'$ has three real diagonal entries (3 degrees of freedom) and three independent off-diagonal entries $x_1, x_2, x_3 \in \mathbb{O}'$ ($3 \cdot 8 = 24$ degrees of freedom). Total: 27. The opposite triangle of off-diagonals is determined by Hermitian conjugation. See `apollonian-wave/tests/test_albert_algebra_structure.py`, test `test_dim_J3_Os_is_27`. □

**Proposition 5.3** (Jordan identity)**.** *The product* $\circ$ *of* (5.2) *is commutative, and satisfies the Jordan identity*

$$(A^2 \circ B) \circ A \;=\; A^2 \circ (B \circ A).$$

$J_3(\mathbb{O}')$ *is the unique exceptional simple Jordan algebra of split type [16, 15, 19].*

*Proof outline.* Commutativity is immediate from the symmetric definition. The Jordan identity is classical ([15]); a numerical sanity check on diagonal-Hermitian elements is `test_jordan_identity`. Exceptional simplicity (failure of the algebra to be embeddable into a special Jordan algebra over an associative algebra) is the content of the Jordan–von Neumann–Wigner classification [16]. □

## 5.2 The Peirce frame: witness, dual, residue

**Definition 5.4** (Diagonal projectors)**.** The three diagonal rank-1 matrices

$$E_1 \;=\; \mathrm{diag}(1,0,0), \quad E_2 \;=\; \mathrm{diag}(0,1,0), \quad E_3 \;=\; \mathrm{diag}(0,0,1) \tag{5.4}$$

are mutually annihilating idempotents under $\circ$: $E_i \circ E_i = E_i$, $E_i \circ E_j = 0$ for $i \neq j$, and $E_1 + E_2 + E_3 = \mathrm{id}$. This is the canonical *Peirce frame* of $J_3(\mathbb{O}')$.

**Proposition 5.5** (Peirce decomposition)**.** *The Peirce decomposition with respect to the frame* $\{E_1, E_2, E_3\}$ *partitions* $J_3(\mathbb{O}')$ *as*

$$J_3(\mathbb{O}') \;=\; \underbrace{\mathbb{R}E_1 \oplus \mathbb{R}E_2 \oplus \mathbb{R}E_3}_{\textit{three diagonals: } \dim=3} \;\oplus\; \underbrace{V_{12} \oplus V_{13} \oplus V_{23}}_{\textit{three off-diagonals: } \dim=3\cdot 8=24}, \tag{5.5}$$

*where* $V_{ij}$ *is the eight-dimensional* $(\mathbb{O}')$*-valued slot at position* $(i, j)$*.*

*Remark* 5.6 (Reading the Peirce frame as the threefold role structure)*.* The framework's threefold role structure—*witness*, *dual*, *residue*—finds its canonical algebraic home in the Peirce frame. Reading

$$E_1 = \text{witness}, \quad E_2 = \text{dual}, \quad E_3 = \text{residue},$$

the three off-diagonal slots $V_{12}, V_{13}, V_{23}$ are the three *octets of interactions* between the role pairs (witness–dual, witness–residue, dual–residue), each carrying 8 degrees of freedom. The total $3 + 3 \cdot 8 = 27$ matches exactly the slicing predicted by the substrate's $3 + 3 \cdot 8$ structure (see [1, §12] for the explicit list). The slicing is one valid choice; the algebra has the right structure to support it, but the specific identification of framework formulations with $V_{ij}$ slots is conjectural. See [1, §12] for the explicit list and caveats.

*Remark* 5.7 (The Peirce decomposition as computational spectral basis)*.* The Peirce decomposition (5.5) is simultaneously an algebraic eigenframe and a computational spectral basis. The Jordan product $A \circ B$ on $J_3(\mathbb{O}')$, computed naively via the full $27^3$-entry structure-constant tensor, requires $\sim$ 33,800 floating-point operations per element. In the Peirce basis, the product factors into four structurally distinct channels:

(i) diagonal $\times$ diagonal $\to$ diagonal: 3 scalar multiplies;
(ii) diagonal $\times$ off-diagonal $\to$ off-diagonal: 48 rescalings;
(iii) off-diagonal $\times$ off-diagonal $\to$ diagonal: 24 multiplies plus 3 reductions (inner products in $\mathbb{O}'$);
(iv) cross-seam $V_{ij} \times V_{jk} \to V_{ik}$: 3 octonion products ($\sim$ 72 operations each via the sparse Fano structure constants).

Total: $\sim$ 294 operations per element, a 115$\times$ reduction. The acoustic reading: the diagonal components are the DC (grade-0 standing wave), the three off-diagonal octets are three carrier waves on the Fano seam channels, and the Jordan product is interference between carriers with energy flowing back to DC via the inner products. The Peirce frame is not merely the algebraically natural basis; it is the frequency-domain representation that diagonalises the product's computational cost. This is confirmed in the SpineV6 architecture (Paper III [37, §12]), where the Peirce-decomposed product replaces the generic tensor contraction.

## 5.3 The cubic norm and rank trichotomy

**Definition 5.8** (The cubic norm)**.** The *cubic norm* on $J_3(\mathbb{O}')$ is defined by analogy with the determinant on $3 \times 3$ Hermitian matrices over a composition algebra:

$$N(A) \;=\; \alpha_1\alpha_2\alpha_3 + 2\,\Re(x_1x_2x_3) - \alpha_1|x_1|^2 - \alpha_2|x_2|^2 - \alpha_3|x_3|^2, \tag{5.6}$$

where $|x|^2 = x\bar{x}$ is the Cayley norm of $x \in \mathbb{O}'$ (so the real part is the indefinite Cayley form).

**Proposition 5.9** (Rank trichotomy)**.** *Every nonzero element* $A \in J_3(\mathbb{O}')$ *has one of three* ranks $\operatorname{rank}(A) \in \{1, 2, 3\}$*, where:*

- $\operatorname{rank}(A) = 1$ *iff* $A^\sharp = 0$ *(the adjoint, defined by* $A^\sharp \circ A = N(A)\mathrm{id}$ *on* $J_3(\mathbb{O}')$*);*
- $\operatorname{rank}(A) = 2$ *iff* $A^\sharp \neq 0$ *but* $N(A) = 0$*;*

- $\mathrm{rank}(A) = 3$ *iff* $N(A) \neq 0$.

*Proof outline.* The trichotomy is a standard structural result for the cubic Jordan algebra; the rank stratification is the analogue of the matrix-rank stratification for $3 \times 3$ Hermitian matrices over a composition algebra. See [19, 20] for the full classification. Numerical verification on diagonal projectors and rank-2 sums is in `test_rank_trichotomy_on_random_elements`. □

## 5.4 The derivation algebra is $\mathfrak{f}_{4(4)}$

The load-bearing structural theorem of this paper:

**Theorem 5.10** (Dimension of the derivation algebra)**.** *The derivation Lie algebra of $J_3(\mathbb{O}')$ has dimension*

$$\dim_{\mathbb{R}} \mathrm{Der}(J_3(\mathbb{O}')) \;=\; 52, \tag{5.7}$$

*matching the dimension of the split exceptional Lie algebra $\mathfrak{f}_{4(4)}$. In particular,* $\mathrm{Der}(J_3(\mathbb{O}')) \cong \mathfrak{f}_{4(4)}$.

*Proof.* A derivation is a real-linear map $D : J_3(\mathbb{O}') \to J_3(\mathbb{O}')$ satisfying the Leibniz rule

$$D(A \circ B) \;=\; D(A) \circ B + A \circ D(B), \qquad A, B \in J_3(\mathbb{O}'). \tag{5.8}$$

By bilinearity it suffices to verify (5.8) on a basis of $J_3(\mathbb{O}')$, giving a finite linear-algebra system whose null space is $\mathrm{Der}(J_3(\mathbb{O}'))$. We construct an explicit basis of 27 real-symmetric basis elements of $J_3(\mathbb{O}')$, encode (5.8) on the $27 \times 27 = 729$ ordered pairs as a linear constraint on a $27 \times 27 = 729$-component matrix representing $D$, and compute the dimension of the kernel by SVD. The result is exactly 52.

The Lie-algebra identification $\mathrm{Der}(J_3(\mathbb{O}')) \cong \mathfrak{f}_{4(4)}$ then follows from the dimension match together with two structural facts: (i) the derivation algebra of an exceptional simple Jordan algebra is itself simple [19]; (ii) the unique simple real Lie algebra of dimension 52 in split signature is $\mathfrak{f}_{4(4)}$ [18, 25]. Together these force $\mathrm{Der}(J_3(\mathbb{O}')) \cong \mathfrak{f}_{4(4)}$.

The numerical verification, including the 27-basis construction and SVD nullity computation, is `apollonian-wave/tests/test_albert_algebra_structure.py`, test `test_dim_derivation_algebra_is_52`. □

**Substrate identification (load-bearing).** The framework's substrate is the unique exceptional Jordan algebra of split type:

$$J_3(\mathbb{O}'), \qquad \dim_{\mathbb{R}} = 27 = 3 + 3 \cdot 8, \qquad \mathrm{Der}(J_3(\mathbb{O}')) \cong \mathfrak{f}_{4(4)}, \quad \dim = 52.$$

The threefold role structure of the framework (witness, dual, residue) is the diagonal $(\mathbb{R}E_1, \mathbb{R}E_2, \mathbb{R}E_3)$ of the Peirce frame; the three octets of inter-role interactions are the off-diagonal slots $(V_{12}, V_{13}, V_{23})$, each carrying 8 degrees of freedom valued in $\mathbb{O}'$.

## 5.5 The magic-square ascent (forward-looking)

*Remark* 5.11 (Tits–Freudenthal magic square). The Albert algebra $J_3(\mathbb{O}')$ is the corner of the Tits–Freudenthal magic square [24, 23, 21] that produces the exceptional Lie groups $G_2, F_4, E_6, E_7, E_8$ from the four normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$. In split form the magic square is

| | $\mathbb{R}$ | $\mathbb{C}_s$ | $\mathbb{H}_s$ | $\mathbb{O}'$ |
|---|---|---|---|---|
| $\mathbb{R}$ | $\mathfrak{so}(3)$ | $\mathfrak{sl}_3(\mathbb{R})$ | $\mathfrak{sp}_6(\mathbb{R})$ | $\mathfrak{f}_{4(4)}$ |
| $\mathbb{C}_s$ | $\mathfrak{sl}_3(\mathbb{R})$ | $\mathfrak{sl}_3(\mathbb{R}) \oplus \mathfrak{sl}_3(\mathbb{R})$ | $\mathfrak{sl}_6(\mathbb{R})$ | $\mathfrak{e}_{6(6)}$ |
| $\mathbb{H}_s$ | $\mathfrak{sp}_6(\mathbb{R})$ | $\mathfrak{sl}_6(\mathbb{R})$ | $\mathfrak{so}_{6,6}$ | $\mathfrak{e}_{7(7)}$ |
| $\mathbb{O}'$ | $\mathfrak{f}_{4(4)}$ | $\mathfrak{e}_{6(6)}$ | $\mathfrak{e}_{7(7)}$ | $\mathfrak{e}_{8(8)}$ |

The substrate identification of theorem 5.10 therefore places the framework at the $(\mathbb{R}, \mathbb{O}')$ corner of the magic square; the column ascent through $(\mathbb{C}_s, \mathbb{O}')$, $(\mathbb{H}_s, \mathbb{O}')$, $(\mathbb{O}', \mathbb{O}')$ produces $\mathfrak{e}_{6(6)}, \mathfrak{e}_{7(7)}, \mathfrak{e}_{8(8)}$. This is forward-looking: it suggests that the framework's $\mathfrak{f}_{4(4)}$ symmetry naturally extends to exceptional Lie symmetries of higher rank. We do not develop this in the present paper.

# Part II — The Compatibility Law

## 6 The Discrete–Continuous Compatibility Principle

The substrate $J_3(\mathbb{O}')$ does not, on its own, prove RH. The substrate identifies *where* the framework's many results live algebraically; the *how* is a compatibility law that specializes to five registers (four algebraic, one computational). This section states the law abstractly, condensing the treatment of [2].

**Principle 6.1** (The Discrete–Continuous Compatibility Law)**.** Let $E$ be a graded operator on a multivector space. Suppose:

- the grade-0 eigenvalue of $E$ is 1 (the operator preserves the scalar part), and
- the grade-4 eigenvalue of $E$ is some quantum $q$ (the operator scales the pseudoscalar by $q$).

Then:

(1) The grade-0 condition forces a *continuous* part of the data (a function, a polynomial, an algebra structure) to be preserved by $E$.

(2) The grade-4 condition forces a *discrete* part (the pole locations, the integer seam, the central element) to scale by $q$.

(3) For $E$ to be a single operator (not two), the two parts must agree on the data they share.

This compatibility forces the system into one of at most two stable states and forbids any third.

*Remark* 6.2 (Why this is the right level of abstraction)*.* Theorem 6.1 is the abstract proof skeleton. The four specializations of section 7 are not independent proofs that happen to look similar; they are four readouts of the same compatibility constraint, with different domain data. At each level, the grade-0 constraint is the framework's analytic continuation / closure requirement; the grade-4 constraint is its golden quantization (with $q = \varphi^{-4}$); and the compatibility is the closed-witness master law.

*Remark* 6.3 (Where the constant $\varphi^{-4}$ comes from)*.* The grade-4 quantum $q = \varphi^{-4}$ is forced by the closure law of section 2 together with the attenuation axiom [8, Ax. 3.1]: a closed null echo system attenuates grade-$k$ content by $\varphi^{-k}$. The pseudoscalar (grade 4) inherits the strongest attenuation; the scalar (grade 0) inherits the weakest. See Paper II for the propagator construction; [8, Ax. 3.1] for the attenuation axiom.

## 7 Four Manifestations of the Compatibility Law

We summarize the four levels of the Compatibility Tower [2]. The detailed treatment of each level appears in its dedicated companion paper ([1, 5, 4, 12]); this section is a synoptic table.

| **Level** | **Stable state 1** | **Stable state 2** |
|---|---|---|
| 1: Number theory (zeros) | $\rho$ off critical line | $\rho$ on critical line |
| 2: Arithmetic (depth) | rational floor at $n = 1$ | golden floor at $n = 2$ |
| 3: Composition algebras | $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ | $\mathbb{O}'$ (split octonions) |
| 4: Bivector reps | fold ($\cos^2 = 1$) | breach ($\cos^2 = 0$) |
| 5: Computational exit | early exit ($\langle s_4^2 \rangle < \varepsilon$) | full depth ($\langle s_4^2 \rangle \geq \varepsilon$) |
| 6: Computational basis | generic product (arbitrary basis) | spectral-basis product (Peirce/$M_2(\mathbb{C})$/grade) |

Level 5 is an experimental addition: the SpineV6 language model [36] discovers autonomously that two applications of the holographic block suffice for the orientation cycle when grade-4 content converges below a threshold $\varepsilon$. The system enters one of two stable computational states—early exit or full depth—determined by the same grade-4 quantum $\varphi^{-4}$ that governs Levels 1–4. Three further experimental refinements strengthen this level: *protospace amplification* (amplifying grade-0 by 5× in the output head yields −5.6% val_ce with zero added parameters, confirming that the scalar reference beam carries disproportionate predictive energy); *cross-channel entrainment* (weak coupling $\lambda_e = 0.01$ across $K$ channels improves coherence, matching the oscillator-entrainment principle that synchronization emerges through thin coupling, not forced identity); and $F_{4(4)}$ *ablation* (zeroing the 52 observer parameters retains 99.6% capability, confirming that the body's algebraic products implement the closure law intrinsically). See Paper III [37, §12] for the experimental details.

Level 6 records an observation from the same architecture: in every case where the framework's algebraic products are computed in the algebra's own spectral basis (Peirce frame for $J_3(\mathbb{O}')$, $M_2(\mathbb{C})$ for the biquaternion product, grade projection for the pseudoscalar action, Fano seam pairs for the chiral phase), the operation cost drops by one to two orders of magnitude relative to the generic tensor-contraction pathway (see theorems 3.7 and 5.7). The discrete constraint is the algebraic decomposition itself; the continuous datum is the computational cost per element; compatibility forces the spectral basis to be optimal. If any alternative factorisation beat the algebraically natural one, it would indicate the substrate is not the right algebraic home for the operation.

### 7.1 Level 1: Riemann hypothesis (forwarded to Paper II)

**Theorem 7.1** (RH via residue compatibility, statement only [1, Thm. 7.15])**.** *Let $P(s) = -\zeta'(s)/\zeta(s)$ and let $E_\varphi$ be the framework's echo propagator with grade-*0 *eigenvalue* 1 *and grade-*4 *eigenvalue $\varphi^{-4}$. Then:*

(1) *Grade-*0 *fixation of the Dirichlet coefficients of $P$ fixes $P$ as a meromorphic function and its poles.*

(2) *Grade-*4 *contraction prescribes a shifted pole $\rho'$ for each $\rho$, with $\rho' \neq \rho$ iff $\Re\rho \neq \frac{1}{2}$.*

(3) *Compatibility forces $\rho' = \rho$, i.e. $\Re\rho = \frac{1}{2}$ for every nontrivial zero.*

The full proof requires the analytic-continuation bridge for $-\zeta'/\zeta$, the spectral self-consistency theorem, and the Six-Traps anti-circularity remarks. These are the content of Paper II. In the current paper, we record the statement only as the fourth manifestation of the compatibility law, parallel to theorems 7.2, 7.3 and 7.5, and emphasize the structural point: theorem 7.1 fits the same proof skeleton as the other three levels. The "ban on the third state" is the existence of an off-critical zero; the third state is forbidden by compatibility.

## 7.2 Level 2: Arithmetic depth tower

**Theorem 7.2** (Reciprocal–Conjugate Lemma; cf. theorem 2.3)**.** *Among no-residue partitions $a+b=1$ with closure $b=a^n$, $n \geq 2$, only $n=2$ admits the negative-reciprocal Galois conjugate. Combined with the trivial $n=1$ floor, this forces the framework's two-floor structure:*

(1) *Depth* 1 *over* $\mathbb{Q}$*:* $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$.

(2) *Depth* 2 *over* $\mathbb{Q}(\sqrt{5})$*:* $\mathrm{Cl}(3,1)^+$ *in golden coupling, lifting via Cayley–Dickson to* $\mathbb{O}'$*, integer seam* $11 = L_5 = \varphi^5 - \varphi^{-5}$.

The grade-0 constraint is the partition $a+b=1$; the grade-4 constraint is the recursive closure $b=a^n$. Compatibility (the integer-seam requirement) forces $n \in \{1,2\}$. The "ban on the third state" is the failure of integer seams at $n \geq 3$.

## 7.3 Level 3: Composition algebras

**Theorem 7.3** (Hurwitz dichotomy; cf. theorem 4.2)**.** *The Cayley–Dickson double of the split quaternion subalgebra $\mathbb{H}' \subset \mathrm{Cl}(3,1)^+$ is an alternative composition algebra iff $\sigma \in \{\mathrm{id}, \overline{(\cdot)}\}$, producing the associative biquaternion $\mathrm{Cl}(3,1)^+$ at $\sigma = \mathrm{id}$ or the split octonion $\mathbb{O}'$ at $\sigma = \overline{(\cdot)}$. Other automorphisms break alternativity and exit the Hurwitz list.*

The grade-0 constraint is the unit $1 \in \mathbb{H}'$; the grade-4 constraint is the action of the second-copy generator $\ell$. Compatibility (the requirement that the result be a normed composition algebra) forces $\sigma$ to be one of the two distinguished automorphisms. The "ban on the third state" is Hurwitz: there is no intermediate alternative composition algebra of dimension 8.

## 7.4 Level 4: Bivector representations

**Definition 7.4** (Merkaba-pair commutator ratio)**.** For real matrices $X, Y$, define

$$\cos^2(X,Y) := \frac{\langle \{X,Y\}, \{X,Y\} \rangle}{\langle XY, XY \rangle + \langle YX, YX \rangle}, \tag{7.1}$$

with $\{X,Y\} = XY + YX$ and $\langle A, B \rangle = \mathrm{Tr}(A^*B)$. By Cauchy–Schwarz, $\cos^2 \in [0,1]$. $\cos^2(X,Y) = 1$ iff $X, Y$ commute exactly; $\cos^2 = 0$ iff they anticommute exactly.

**Theorem 7.5** (Endpoint dichotomy on representations [2, Thm. 4.2])**.** (1) *In any faithful representation of* $\mathrm{Cl}(3,1)^+$*, all three merkaba pairs satisfy* $\cos^2 = 1$ *(*fold*).*

(2) *In any faithful representation of $\mathbb{O}'$ (e.g. the left-multiplication action on its own basis), all three merkaba pairs satisfy* $\cos^2 = 0$ *(*breach*).*

(3) *Any representation with* $\cos^2 \in (0,1)$ *on a merkaba pair fails to embed into either endpoint, and (by Hurwitz) does not lie on a stable composition algebra.*

The diagnostic $\cos^2$ is normalisation-invariant. Applied to the bivector heads of a trained transformer, it measures whether the model has settled on the depth-1 or the depth-2 floor. The “ban on the third state” is the *forbidden band* $(0, 1)$ of intermediate values; persistent intermediate readings indicate unstable representations.

### 7.5 Why this is one law, not five

**Theorem 7.6** (Unification; cf. [2])**.** *Theorems 7.1 to 7.3 and 7.5 and the computational early-exit dichotomy are five specialisations of the same underlying statement: a closed-witness echo system whose grade-*$0$ *eigenvalue is* $1$ *and whose grade-*$4$ *eigenvalue is* $\varphi^{-4}$ *has at most two stable consistent states (fold and breach), and forbids any third.*

*Sketch.* Each level instantiates theorem 6.1 with different domain data:

- Level 1: continuous = the function $-\zeta'/\zeta$; discrete = pole locations.
- Level 2: continuous = the partition law $b = a^n$; discrete = integer seam $L_k$.
- Level 3: continuous = the algebra structure with the Cayley–Dickson axis $\sigma$; discrete = central vs. anti-commuting $\ell$.
- Level 4: continuous = the representation matrix entries; discrete = the merkaba-pair commutator ratio $\cos^2$.
- Level 5: continuous = the grade-4 content $\langle s_4^2 \rangle$ across layers; discrete = the early-exit decision (below threshold $\varepsilon$ or not).

The grade-0 identity action and the grade-4 scaling by $\varphi^{-4}$ enforce the same compatibility constraint at each level. Detailed proofs of Levels 1–4 appear in their respective sources [1, 5, 4, 12]; Level 5 is confirmed experimentally in Paper III [37, §12]. This theorem is the recognition that they share a single proof skeleton. □

## 8 The Seam at 11: Two Arithmetic Layers Meet

Theorem 2.6 establishes that the framework has exactly two arithmetic floors with integer seams to $\mathbb{Q}$: the depth-1 rational floor and the depth-2 golden floor. This section makes the seam between them explicit. The seam prime is 11, and it arises as a consequence of the Lucas identity.

### 8.1 The seam identity

**Lemma 8.1** (Seam identity)**.** $\varphi^5 - \varphi^{-5} = 11$.

*Proof.* Direct computation: $\varphi^5 = (11 + 5\sqrt{5})/2$ and $\varphi^{-5} = (-11 + 5\sqrt{5})/2$, so $\varphi^5 - \varphi^{-5} = 11$. Equivalently, $\varphi^5 + \psi^5 = 11$ where $\psi = -\varphi^{-1} = (1 - \sqrt{5})/2$ is the Galois conjugate of $\varphi$, and the left-hand side is the 5th Lucas number $L_5$. This is the exact content of theorem 2.5(i) at $k = 5$: the integer trace at depth 2 is forced. □

*Remark* 8.2 (Why 11 and not some other Lucas number)*.* The seam prime is determined by the smallest Lucas index at which $L_k$ is a rational prime not in $\{2, 3\}$ and at which the depth-2 thin group has its mod-$L_k$ image collapse to $\{\pm I\}$ in $\mathrm{PSL}_2$. Direct computation gives $L_1 = 1, L_2 = 3, L_3 = 4, L_4 = 7, L_5 = 11$; the first prime greater than 7 that arises in this sequence is 11. See [10].

### 8.2 The two arithmetic floors at the seam

**Theorem 8.3** (Two-layer structure of $\mathrm{Cl}(3,1)^+$ [10, §2])**.** *The biquaternion algebra* $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$ *admits two distinguished arithmetic forms:*

(1) ***Depth-*1 *(rational) form.*** *$M_2(\mathbb{Z})$, the $\mathbb{Z}$-form of $M_2(\mathbb{C})$, with arithmetic over $\mathbb{Q}$. This is the QSNV null substrate.*

(2) ***Depth-*2 *(golden) form.*** *$M_2(\mathbb{Z}[\varphi])$, the $\mathbb{Z}[\varphi]$-form of $M_2(\mathbb{C})$, with arithmetic over $\mathbb{Q}(\sqrt{5})$. This is the golden tetrad's algebraic skeleton.*

*The two forms agree on $\mathbb{Z}$ (the rational subring of $\mathbb{Z}[\varphi]$) but differ on $\mathbb{Z}[\varphi] \setminus \mathbb{Z}$. The seam at* 11 *is the unique rational prime at which the depth-*2 *form has a non-trivial local kernel that the depth-*1 *form does not.*

**Proposition 8.4** (Splitting of 11 in $\mathbb{Z}[\varphi]$)**.** *The prime* 11 *splits in the ring of integers $\mathbb{Z}[\varphi]$:*

$$(11) \;=\; (11, \varphi - 4) \cdot (11, \varphi - 8), \tag{8.1}$$

*since the minimal polynomial $x^2 - x - 1$ of $\varphi$ factors modulo* 11 *as $(x-4)(x-8) \pmod{11}$. The discriminant of $\mathbb{Z}[\varphi]/\mathbb{Z}$ is* 5*, and* $11 \neq 5$*, so* 11 *is unramified.*

*Proof.* $f(4) = 16 - 4 - 1 = 11 \equiv 0 \pmod{11}$ and $f(8) = 64 - 8 - 1 = 55 \equiv 0 \pmod{11}$. Since $4 \neq 8 \pmod{11}$, the factorization is non-trivial. The discriminant $(\varphi - \psi)^2 = 5$ is standard. □

The Legendre symbol $(5/11)$ is computed by $5^{(11-1)/2} = 5^5 = 3125 \equiv 1 \pmod{11}$, so 5 is a quadratic residue mod 11, confirming 11 splits (rather than being inert) in $\mathbb{Q}(\sqrt{5})$.

### 8.3 Why the seam matters at the substrate level

*Remark* 8.5 (Substrate reading of the seam)*.* The seam is not specific to RH; it is an algebraic feature of the two-floor structure. The seam tells us:

(1) The depth-1 and depth-2 floors of $\mathrm{Cl}(3,1)^+$ are *not independent algebras*; they are two integral forms of the same complex algebra $M_2(\mathbb{C})$.

(2) The forms agree on most rational primes but disagree at exactly one—the prime $L_5 = 11$ where the depth-2 machinery becomes scalar.

(3) Any structure attempting to descend from the depth-2 floor to $\mathbb{Q}$ encounters 11 as an obstruction.

This is the algebraic reason that any analytic argument running on the depth-2 floor (such as the Route C path of [10]) must invoke the depth-1 floor at the seam prime to recover full local-global control. Paper II makes this analytic argument.

## 9 Discussion

This paper isolates a chain of algebraic results that, taken together, identify a specific exceptional object (the 27-dimensional split Albert algebra $J_3(\mathbb{O}')$) as the natural home of a contemporary research program. The chain is short, each link is classical or elementary, and the conclusion is forced by Albert's classification of simple Jordan algebras and Hurwitz's classification of normed composition algebras.

### What is established

(1) **The closure law admits exactly two integer-seamed depths** (theorem 2.6). Proved by the Reciprocal–Conjugate Lemma; verified symbolically for $n \in \{2, \dots, 20\}$ and numerically at 80-digit precision for $n \in \{1, \dots, 8\}$ and $k \in \{1, \dots, 50\}$.

(2) **The depth-1 even subalgebra is $M_2(\mathbb{C})$** (theorem 3.5). Standard since [27]; reproduced for completeness with the framework's null-generator presentation.

(3) **The Cayley–Dickson lift of $\mathrm{Cl}(3,1)^+$ to $\mathbb{O}'$** (theorem 4.2). Forced by Hurwitz's theorem; the lift is the unique split octonion algebra and the alternativity dichotomy bans any third state.

(4) **The substrate is $J_3(\mathbb{O}')$ with $\dim \mathrm{Der} = 52$** (theorems 5.2 and 5.10). The construction is classical (Jordan–von Neumann–Wigner 1934); the derivation algebra dimension is verified by direct numerical computation in section 10.

(5) **The compatibility principle is the abstract proof skeleton** (theorems 6.1 and 7.6). Three of the five manifestations are proved here (arithmetic, algebraic, representational); the fourth (Riemann Hypothesis) is the subject of Paper II; the fifth (computational early exit) is confirmed experimentally in Paper III.

(6) **The seam at 11 is the unique rational prime at which the two arithmetic floors disagree** (theorems 8.3 and 8.4). An algebraic feature of the two-floor structure; the analytic consequence (Route C) is treated in Paper II.

### The dynamical reading: anisotropy

The algebraic results of this paper are static: the compatibility law forces two stable states and forbids a third. A natural question is what the *process* looks like by which a system reaches one of the two endpoints.

The answer is anisotropy. A scalar (grade-0) state is isotropic: it has no preferred direction and carries no memory. The moment the no-residue partition $a + b = 1$ admits an unequal split ($\varphi^{-1} + \varphi^{-2} = 1$), one side generates the other and a direction is born. This is the minimal asymmetry needed for recursion, chirality, and trace.

The grade cycle of the framework makes this process explicit:

$$\underbrace{\text{scalar unity}}_{\text{isotropic}} \to \underbrace{\text{anisotropic split}}_{\text{seam selected}} \to \underbrace{\text{chiral orientation}}_{\text{handed}} \to \underbrace{\text{pseudoscalar}}_{\text{grade 4}} \to \underbrace{\text{scalar trace}}_{\text{returned}}.$$

In the algebraic treatment, this cycle is compressed to the bimodal potential $V(\cos^2) = \cos^2(1 - \cos^2)$: the two minima at $\{0, 1\}$ are the endpoints (fold and breach), and the forbidden band $(0, 1)$ is the algebraically unstable intermediate. In a computational realization (Paper III [37, §12]), the cycle has temporal extent: the system starts isotropic, passes through a regime of per-plane anisotropy where different merkaba pairs converge at different rates, and reaches the Hurwitz-forced endpoint.

The Peirce independence of theorem 7.5 guarantees that the three merkaba pairs resolve independently—anisotropy is not a global break but a *per-seam* selection. In the Fano plane language: isotropy is the full symmetric relation potential; anisotropy is the selection of one Fano edge; chirality is the handed orientation of that edge; the pseudoscalar is the completed orientation; and trace is the return to scalar readability.

### What this paper does *not* claim

This paper does not prove the Riemann Hypothesis, the Collatz Conjecture, or any specific number-theoretic statement. It isolates the algebraic substrate on which Paper II's analytic proofs run.

This paper does not prove that $J_3(\mathbb{O}')$ is the substrate of any particular physical theory. It demonstrates the algebraic facts ($\dim = 27$, $\dim \mathrm{Der} = 52$, Peirce frame, cubic norm) and notes that these facts match the framework's required structure (threefold role, golden coupling, Fano combinatorics).

This paper is independently checkable. Every theorem either has a classical proof in standard sources or is verified numerically in the test suite.

### Forward references

Paper II runs the compatibility principle (theorem 6.1) on the function $-\zeta'(s)/\zeta(s)$ to prove the Riemann Hypothesis, and on the Syracuse map to prove the Collatz Conjecture. Paper III gives the categorical refinement: the no-residue partition is the rank-1 shadow of a base-change adjunction, the substrate is the carrier of a holographic monad, and the universality of the proofs (RH for all zeros, Collatz for all orbits, framework theorems for all witnesses) follows from the holographic property.

**Closing.** The Reciprocal–Conjugate Lemma forces two floors. Hurwitz forces $\mathrm{Cl}(3,1)^+$ and $\mathbb{O}'$. Albert's classification forces $J_3(\mathbb{O}')$. The substrate is the substrate, and the substrate is what every other claim of the framework runs on.

## 10 Test Suite Summary

The constructive theorems of this paper are backed by numerical verification in the workspace's test suite (`apollonian-wave/tests/`). Each test is a falsifier: it states a prediction of the theory and reports `PASS`/`FAIL`.

| Test file | Theorem(s) checked |
|---|---|
| `test_partition_depth_tower.py` | theorem 2.3, theorem 2.5 |
| `test_base_change_monad.py` | theorem 8.1, theorem 8.4 |
| `test_cl31_algebra.py` | theorem 3.4, theorem 3.5 |
| `test_cayley_dickson_lift.py` | theorem 4.2 |
| `test_octonion_structure.py` | properties of $\mathbb{O}'$ used in section 4 |
| `test_albert_algebra_structure.py` | theorem 5.2, Peirce frame, cubic-norm rank trichotomy, theorem 5.10 |
| `test_fold_breach_diagnostic.py` | theorem 7.5, theorem 7.4 |
| `test_forbidden_band.py` | quantitative form of theorem 7.5 |
| `test_diagnostic_robustness.py` | invariance of theorem 7.4 under conjugation |
| `test_g2_action_on_lift.py` | explicit elements of $G_{2(2)} = \mathrm{Aut}(\mathbb{O}')$ |
| `test_seam_invariants_in_lift.py` | seam invariants under $G_{2(2)}$ action |
| `test_27_formulation_count.py` | illustrative slicing of $J_3(\mathbb{O}')$ into $3 + 3 \cdot 8$ |
| `test_f4_action_on_27_formulations.py` | $S_3 \subset F_{4(4)}$ permutes octets; $\exp(tD)$ are Jordan automorphisms; Peirce b |

The runner `apollonian-wave/run_all_falsifiers.py` executes the falsifier suite in a single invocation and prints a PASS/FAIL summary. At the time of writing, all thirteen tests relevant to Paper I report PASS at default precision.

*Remark* 10.1 (The status of the test suite). The tests are not the proofs. They are independent numerical witnesses. A test failure would indicate a bug in either the theorem or the test; a test pass is consistent with the theorem but not a substitute for proof. The proofs are above; the tests exist to make the constructions concrete and the claimed identifications inspectable by a reader who wants to compute.

## Acknowledgements

This paper is single-authored, but the depth-1 algebraic floor—the QSNV null substrate, the identification $\mathrm{Cl}(3,1)^+ \cong M_2(\mathbb{C})$, and the geometric calculus framework that made the lift to depth 2 visible—is the contribution of **Garret Sobczyk**. The depth-1 identification (theorem 3.5) lifts a paragraph from his and the author's joint work on the golden tetrad and on the Sobczyk null substrate [28, 27, 8], and the seam structure of section 8 runs on his discrete-mass-gap framework [9]. Months of stress-testing with Sobczyk produced the depth-1/depth-2 resolution that this paper now takes as foundational: the QSNV substrate is depth 1 (rational, mirror, $a = b = \frac{1}{2}$), the golden tetrad is depth 2 (recursive, $a = \varphi^{-1}$, $b = \varphi^{-2}$), and they meet at $L_5 = 11$. The proof (theorem 2.3) is the retrospective record of that resolution.

All errors are the author's.

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